FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 · SUB-STRAND 3.1: TRIGONOMETRY I	
Student Assessment Document	
Student Name	______________________________
Class	______________________________
Date	______________________________

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation — your opportunity to show everything you have learned across our 10-lesson Trigonometry I sequence. You have investigated the anchoring phenomenon: a student in Kisumu used her own shadow and the shadow of a mango tree, plus a clinometer angle of 53°, to figure out the tree's height without climbing it. Now it is your turn to explain, clearly and completely, how that is possible.

Read each section prompt carefully. Write in full sentences. Use correct mathematical vocabulary (trigonometric ratio, sine, cosine, tangent, opposite, adjacent, hypotenuse, angle of elevation, similar triangles). Show all working where calculations are required. Your explanation should be detailed enough that a fellow student who missed every lesson could read your work and fully understand how angles unlock unknown heights. Good luck!

KICD Learning Outcomes addressed: Apply trigonometric ratios to solve problems involving right-angled triangles; establish relationships among sine, cosine, and tangent of acute angles; connect mathematical reasoning to real-world contexts.

Competencies: Communication and collaboration · Critical thinking and problem solving · Creativity and imagination.
Core Values: Responsibility · Integrity · Social justice.
PCIs: Environmental awareness (indigenous trees, land use) · Financial literacy (construction, surveying).

Section 1 — The Hidden Pattern: Similar Triangles and Consistent Ratios

In our opening lesson, a student in Kisumu noticed that her 1.6 m height cast a 1.2 m shadow, while the mango tree's shadow was 9.0 m long — all at the same moment on the same sunny afternoon.	(a) [Diagram shows two right-angled triangles. Triangle 1 — student: vertical side = 1.6 m (height), horizontal side = 1.2 m (shadow), hypotenuse = sun's ray. Triangle 2 — tree: vertical side = h (unknown height), horizontal side = 9.0 m (shadow), hypotenuse = sun's ray. Both triangles share the same sun-ray angle at the base because the sun's rays are parallel at the same moment in time, so the angle of elevation of the sun is identical for both.] (b) Since the triangles are similar (AA similarity — same sun angle, both have a 90° angle), their corresponding sides are proportional:

(a) Draw and label two right-angled triangles: one for the student and one for the mango tree. Mark the height (vertical side), shadow length (horizontal side), and the sun's ray (hypotenuse) on each triangle. Explain why both triangles must contain the same angles. **(b) Using the concept of similar triangles, show how you can calculate the height of the mango tree from the shadow measurements alone. Show your working fully.** **(c) Explain what this tells us about the RATIO of opposite side to adjacent side in both triangles. Why does this ratio stay the same even though the triangles are very different in size?**	Height of student / Shadow of student = Height of tree / Shadow of tree 1.6 / 1.2 = h / 9.0 h = (1.6 × 9.0) / 1.2 h = 14.4 / 1.2 h = 12.0 m The mango tree is 12.0 m tall. (c) In both triangles, the ratio (opposite ÷ adjacent) = (height ÷ shadow length) equals 1.6/1.2 = 4/3 ≈ 1.333. This ratio is the same because the triangles are similar — they have identical angles. Whenever an angle is fixed, the ratio of the opposite side to the adjacent side is always the same, no matter how big or small the triangle is. This consistent ratio is exactly what the tangent trigonometric ratio captures.

Section 2 — Defining the Three Trigonometric Ratios	
You have discovered that for a fixed angle, the ratio of two sides in a right-angled triangle never changes. Mathematicians have named three of these special ratios: sine, cosine, and tangent. **(a) Define sine, cosine, and tangent for an acute angle θ in a right-angled triangle. Use the words 'opposite', 'adjacent', and 'hypotenuse' in your definitions, and write each as a fraction.** **(b) The classmate's clinometer showed the sun's angle of**	(a) For an acute angle θ in a right-angled triangle: – sin θ = opposite / hypotenuse – cos θ = adjacent / hypotenuse – tan θ = opposite / adjacent A helpful memory aid: SOH-CAH-TOA. (b) sin 53° ≈ 0.7986 | cos 53° ≈ 0.6018 | tan 53° ≈ 1.3270. The value tan 53° ≈ 1.3270 means that at a sun elevation angle of 53°, the height of ANY object is approximately 1.327 times longer than its shadow. So if you know the shadow length, you simply multiply by 1.327 to get the height. This is a powerful tool — it works for a tree, a building, a flagpole, or a hill, all without direct measurement. (c) Relationship 1 — tan θ = sin θ / cos θ. This is because (opp/hyp) ÷ (adj/hyp) = opp/adj = tan θ. The hypotenuses cancel. This shows the three ratios are not independent; tangent is always the quotient of sine and cosine. Relationship 2 — Complementary angles: sin θ = cos(90° − θ). For example, sin 53° = cos 37° ≈ 0.7986. This is because the side that is 'opposite' angle θ is 'adjacent' to the complementary angle (90° − θ) in the same triangle. This means every sine value has a matching cosine value for the complementary angle.

elevation was approximately 53°. Using a calculator or trigonometric tables, state the values of sin 53°, cos 53°, and tan 53° (to 4 decimal places). Explain in your own words what the value of tan 53° ≈ 1.3270 tells you about the relationship between the height of any object and its shadow length at that moment. **(c) Identify and explain TWO relationships between sine, cosine, and tangent — for example, what happens when you divide sin θ by cos θ? What do you notice about sin θ and cos(90° − θ)?**	

Section 3 — Solving the Kisumu Tree Problem Using Trigonometry	
Now bring the two approaches together. The student has both the shadow-ratio method AND the clinometer angle of 53°. **(a) Using trigonometry (not the similar-triangle ratio method), set up and solve an equation to find the height of the mango tree given that the angle of elevation of the sun is 53° and the shadow length is 9.0 m. Show all steps clearly.** **(b) Compare your answer from Section 1(b) (similar triangles) with your answer from part (a) above. Are they the same? If there is a small difference, explain why.** **(c) Imagine the school wants to**	(a) The shadow is the adjacent side, the tree height is the opposite side, and the angle of elevation is 53°. tan 53° = opposite / adjacent = height / 9.0 height = 9.0 × tan 53° height = 9.0 × 1.3270 height ≈ 11.94 m ≈ 11.9 m (1 d.p.) (b) The similar-triangle method gave exactly 12.0 m; the trigonometric method gives approximately 11.9 m. The tiny difference arises because the clinometer reading of 53° is an approximation — the true sun angle calculated from the student's measurements is arctan(1.6/1.2) = arctan(1.333) ≈ 53.13°, not exactly 53°. Both methods are consistent; the slight discrepancy is due to rounding the measured angle. In real fieldwork, measurement error is always present, which is why surveyors take multiple readings. (c) tan 62° = height / 15 height = 15 × tan 62° height = 15 × 1.8807 height ≈ 28.2 m The water tower is approximately 28.2 m tall.

find the height of the new water tower being built near the Kisumu school compound. A student stands 15 m from the base of the tower and measures the angle of elevation to the top as 62°. Calculate the height of the water tower. Show full working and state your answer to 1 decimal place.	

Section 4 — Angles of Elevation and Depression in Kenyan Contexts	
Trigonometry is used every day by architects, farmers, and engineers across Kenya — from designing rooftops in Nairobi to planning irrigation channels in the Rift Valley. **(a) Define 'angle of elevation' and 'angle of depression'. Draw a clearly labelled diagram for each, using a Kenyan context of your choice.** **(b) A spotter at the top of a ranger watchtower in Tsavo East National Park looks down at an elephant on flat ground. The angle of depression to the elephant is 34°, and the tower is 18 m tall. How far is the elephant from the base of the tower? Show full working.** **(c) A farmer in Kericho wants to string a wire from the top of a 5.5 m vertical post to a peg in the ground. The wire must make an angle of 40° with the ground. Calculate: (i) the length of the wire needed, and (ii) the horizontal distance from the**	(a) Angle of elevation: the angle measured upward from the horizontal to a line of sight looking UP at an object. Example diagram — observer at ground level in Nairobi looking up at the top of a tall building; the angle between the horizontal ground and the line of sight to the rooftop is the angle of elevation. Angle of depression: the angle measured downward from the horizontal to a line of sight looking DOWN at an object. Example diagram — observer at the top of a cliff at the Kenyan coast looking down at a fishing boat; the angle between the horizontal and the line of sight down to the boat is the angle of depression. Note: angle of elevation from the boat to the cliff top = angle of depression from the cliff top to the boat (alternate interior angles, since horizontal lines are parallel). (b) The tower height is the opposite side; the horizontal distance to the elephant is the adjacent side; the angle of depression = 34°. tan 34° = opposite / adjacent = 18 / distance distance = 18 / tan 34° distance = 18 / 0.6745 distance ≈ 26.7 m The elephant is approximately 26.7 m from the base of the tower. (c) The post is the opposite side (5.5 m); the wire is the hypotenuse; the angle with the ground is 40°. (i) sin 40° = opposite / hypotenuse = 5.5 / wire length wire length = 5.5 / sin 40° = 5.5 / 0.6428 ≈ 8.6 m (ii) cos 40° = adjacent / hypotenuse = horizontal distance / 8.6 OR tan 40° = 5.5 / horizontal distance → horizontal distance = 5.5 / tan 40° = 5.5 / 0.8391 ≈ 6.6 m The farmer needs approximately 8.6 m of wire, and the peg should be placed 6.6 m from the base of the post.

post base to the peg. Show all working.	

Section 5 — Final Reflection: Answering the Driving Question	
You started this unit with a puzzle: how can an angle — something measured with a protractor — tell you the height of a real tree, building, or hill in Kenya without measuring it directly? **(a) Write a full, coherent explanation (at least one well-developed paragraph) that directly answers the driving question: 'How can the angle a sunbeam makes with the ground allow us to calculate the height of any tree, building, or hill in Kenya without measuring it directly?' Your answer must mention: similar triangles, trigonometric ratios (sine, cosine, tangent), angle of elevation, and at least one real Kenyan application.** **(b) Reflect on how your thinking changed from your very first prediction at the start of the unit to what you now know. What was the most surprising or powerful idea you encountered?** **(c) Identify ONE limitation of using shadow-based trigonometry in Kenya, and suggest how a surveyor or student might overcome it.**	(a) When a sunbeam hits the ground at a fixed angle of elevation, it creates a right-angled triangle between the sun's ray, the vertical object (a tree, building, or hill), and the object's shadow on the ground. Because the sun's rays are effectively parallel across a small area, every object at the same location experiences the same sun angle at the same moment. This means all the right-angled triangles formed — whether by a person, a mango tree in Kisumu, or a telecommunications mast in Nairobi — are similar triangles. Similar triangles have identical angles and proportional sides, so the ratio of any two corresponding sides is constant. Trigonometric ratios — sine (opposite/hypotenuse), cosine (adjacent/hypotenuse), and tangent (opposite/adjacent) — are precisely these constant ratios for a given angle. Therefore, if you measure the angle of elevation of the sun and one accessible length (such as the shadow on the ground), you can use the tangent ratio to calculate the unknown height: height = shadow length × tan(angle of elevation). This principle is used by architects designing tall buildings along Nairobi's skyline, by engineers planning bridges over rivers like the Tana, and by foresters estimating the height of indigenous trees in Kakamega Forest — all without ever leaving the ground. (b) [Model answer — student voice]: At the start of the unit, I thought you could only find the height of something by measuring it physically — climbing it, using a ruler, or dropping a rope. I could not see how an angle alone could give you a length. The most powerful idea I encountered was that the RATIO of two sides in a right-angled triangle depends only on the angle, not on how big the triangle is. Once that clicked — especially when we saw that the student's 1.6/1.2 ratio was identical to the mango tree's 12.0/9.0 ratio — I realised that trigonometric ratios are essentially a dictionary: you look up the angle, and the dictionary tells you the exact ratio of the sides. That completely changed how I see angles. (c) One limitation: shadow-based trigonometry only works when the sun is shining and casting a clear shadow — it fails on cloudy days, near dawn or dusk (when the angle is very low and shadows are extremely long, increasing measurement error), or in dense forests like Karura where shadows overlap. A surveyor can overcome this by using a clinometer or theodolite to measure the angle of elevation directly from a known horizontal distance, without relying on shadows at all. This method works at any time of day and in any lighting condition.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Understanding of Similar Triangles and Proportional Reasoning	Clearly explains why both triangles are similar (parallel sun rays → equal angles), correctly sets up and solves the proportion to find the tree height (12.0 m), and articulates precisely why the ratio of opposite to adjacent is constant across all similar triangles. Diagram is fully labelled.	Correctly identifies the triangles as similar and sets up the proportion with mostly correct working. May have a minor arithmetic error but the conceptual reasoning is sound. Diagram is present and mostly labelled.	Attempts to use proportions but makes significant errors in setting up the ratio or solving it. Limited explanation of why the triangles are similar. Diagram is missing or unlabelled.
Accurate Definition and Application of Trigonometric Ratios (SOH-CAH-TOA)	Defines all three ratios (sin, cos, tan) correctly using opposite, adjacent, and hypotenuse. Accurately applies the correct ratio to solve all problems (tree, water tower, wire, elephant) with full working shown and answers stated to the correct degree of accuracy. Correctly identifies and explains both the tan = sin/cos relationship and the complementary angle relationship.	Defines the three ratios correctly and applies them accurately in most problems. May make one computational error or omit one of the two relationships between ratios. Working is shown but may lack a final statement of the answer with units.	Defines one or two ratios but confuses opposite/adjacent, or applies the wrong ratio to a problem. Working is incomplete or difficult to follow. Only one relationship between ratios is identified, or neither is fully explained.
Correct Use of Angles of Elevation and Depression	Defines both terms precisely and supports each with a clearly labelled Kenyan-context diagram. Correctly identifies which side is opposite and which is adjacent in both elevation and depression scenarios. Solves the Tsavo watchtower problem and the Kericho wire problem with accurate working and correct answers (26.7 m; 8.6 m wire; 6.6 m horizontal distance).	Defines both terms correctly and draws diagrams. Solves at least two of the three calculation parts correctly. May confuse angle of elevation and depression in the diagram or misidentify opposite/adjacent in one problem.	Defines only one term, or definitions are confused. Diagrams are missing or incorrectly labelled. Attempts at least one calculation but makes fundamental errors in identifying sides relative to the angle.
Quality of Final Explanation Answering the Driving Question	Writes a well-developed, coherent paragraph that directly and fully answers the driving question. Correctly integrates all required concepts — similar triangles, all three trigonometric ratios, angle of elevation, and a specific Kenyan application. The explanation could genuinely help a peer who missed all lessons. Reflection is thoughtful and shows clear evidence of conceptual change from the initial prediction.	Answers the driving question with most required concepts present. The explanation is coherent but may lack depth on one concept (e.g., similar triangles is mentioned but not fully explained) or the Kenyan application is generic. Reflection shows some evidence of conceptual change.	Attempts to answer the driving question but the explanation is incomplete, disjointed, or missing two or more required concepts. The link between angles, ratios, and real heights is not clearly articulated. Reflection is superficial or absent.
Mathematical Communication:	All working is shown step-by-step for	Working is shown for most calculations.	Working is frequently missing or shows

Working, Notation, and Accuracy	every calculation. Correct mathematical notation is used throughout (e.g., tan 53° = opp/adj, not 'tangent equals height over shadow'). Answers include appropriate units (metres). Rounding is applied correctly and consistently to 1 decimal place where instructed. Diagrams are neat, labelled, and consistent with calculations.	Notation is mostly correct with minor lapses. Units are included in most answers. Some rounding inconsistencies but final answers are largely correct. Diagrams support the work.	only a final answer. Mathematical notation is informal or incorrect. Units are often omitted. Diagrams are absent or do not correspond to the calculations attempted.